and forecasts up to a few hours ahead using radar products.

Short range forecasts is for one to three days. Weather mainly rainfall for each successive 24 hour intervals may be predicted up to three days. It concerned about the observed latest weather charts and new systems.

Medium range forecasts are for 4 to 10 days – average weather conditions and the weather on each day may be prescribed with progressively lesser details and accuracy than that for short range forecasts dependent on NWP (Numerical Weather Prediction) products.

Long range forecast is an extended range forecast for more than 10 days in a season. It may range from a monthly to a seasonal forecast. Local forecast is a location specific forecast valid for a radius of 50 km around the weather stations.

A persistence forecast predicts that future weather will be the same as present weather. Analogue forecast will be like the weather that historically occurred when similar conditions were present. Statistical forecast is made routinely of weather elements based on the past performance of computer models. Trend forecast: surface weather systems tend to move in the same direction and at approximately the same speed as they have moved.

Current Scenario of Weather Forecasting

The weather satellites monitor the weather conditions and provide accurate information of weather. Satellite imageries are pictorial representations of radiation reaching the sensors from the earth from the different spectral bands that imprints the accurate weather elements such as winds, rainfall, and sea surface temperature.

Weather forecasting is based on the weather observation of surface data based on radars. Numerical Weather Prediction (NWP) using current and past observations to predict weather in near future AWIPS- Advanced Weather Interactive Processing Systems used by forecasters and process satellite, radar, surface observations, and weather forecasting models.

Occultation method is one of the most recent and capable atmospheric remote sensing technique applied to GPS measurements.

The meteogram is a chart that shows how one or more weather variables has changed at a station over a given period of time. Auxiliary charts- satellite imageries, satellite bulletins, satellite observations and current weather observations. These predictions are significant for warning of natural hazards.

12.9 Tracking of Cyclones

Tropical cyclone forecasting involves the prediction of several interrelated features, including the track of the cyclone, its intensity, resulting rainfall and storm surge and the areas threatened. Cyclone tracking is a constantly evolving science.

Different methods are used, including using satellites, radars, etc. A skilled meteorologist has often developed an ability to detect overall patterns in climatological conditions and can assess how these may affect cyclone development. Manual forecasts made by skilled

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